GIDFINE — MINI-SCALE IMPLEMENTATION READINESS AUDIT

Reference: Gidfine_Dev_Guide.md (v1.1) | **Active Branch:** development | **Commit:** afb4323 | **Date:** October 21, 2026

Executive Summary: Readiness assessment for implementing the initial mini-scale loan transaction lifecycle (Borrower Origination → Agency Review → Masked Lender Proposal → Agency Approval → Borrower Consent → Disbursement Prerequisites) on React 19 + Convex. Strictly decouples approved specification rules from pending business decisions (Section 15), preserving all existing components and Python backend files.

1. Reusable React Frontend Inventory

Workflow Domain	Component Path	Convex Adaptation Scope
Borrower Origination	workspace/pages/LoanApplication.tsx widgets/OnboardingWidget.tsx	Connect form inputs to Convex originateApplication mutation in integer pesewas.
Agency Review Queue	AdminDashboard.tsx DashboardPane.tsx	Wire review queue to Convex listAgencyApplications query filtering by status.
Lender Matching & Proposal	LoanComparisonTable.tsx LenderManagement.tsx	Display proposal dispatch form; wire to Convex assignLenderProposal mutation.
Lender Review Portal	workspace/widgets/OffersWidget.tsx LoanCard.tsx	Wire to getLenderProposalView query (strictly redacting borrower PII on server).
Borrower Consent / Sanction	workspace/widgets/SanctionViewer.tsx	Replace mock IMPS action with Convex recordBorrowerConsent mutation snapshot.

2. Mini-Scale Loan Workflow & Control Matrix

Stage & Transition	Authorized Role	Required Input Data	Audit & Isolation Invariants
1. Submission DRAFT → SUBMITTED	borrower	Amount (pesewas), tenure (months), purpose	Tagged with verified agencyId; caller must own borrower profile.
2. Agency Review SUBMITTED → MATCHING	agency_admin loan_officer	applicationId, target lenderId, initial terms	Tenant boundary enforced; officers from Agency A cannot view Agency B.
3. Lender Response LENDER_PENDING → LENDER_ACCEPTED	lender	applicationId, decision (accept/decline), rate (bps)	**Act 843 Invariant:** Borrower name/phone/email/address redacted on server.
4. Agency Approval LENDER_ACCEPTED → AGENCY_APPROVED	agency_admin	applicationId, proposalId, approved margin (bps)	Non-bypass invariant: Lender accept cannot substitute for agency sign-off.
5. Borrower Consent AGENCY_APPROVED → BORROWER_ACCEPTED	borrower	applicationId, affirmative consent boolean	Stores immutable snapshot of approved terms; terms hash recorded in audit log.
6. Disbursement Ready BORROWER_ACCEPTED → PENDING_DISBURSEMENT	cashier agency_admin	Disbursement rail ref, verified payee details	Repayment schedule active **only** upon confirmed funding posting.

3. Specification Boundary Classification

Approved Specification (Gidfine_Dev_Guide.md v1.1)	Pending Decision (Section 15 Questions)

• 11-stage linear state transition sequence & control rules • 7 distinct user roles and authority boundaries • Mandatory server-side field redaction for lenders (Act 843) • Append-only audit logs for all status & financial events • All monetary values stored as integer minor units (pesewas)	• Tenancy Model: Dedicated single-agency vs. SaaS multi-tenant • Calculation Standard: Flat simple interest vs. reducing-balance • Margin Mechanism: Rate spread (bps) vs. upfront flat origination fee • Identity Provider: Clerk vs. Auth0 for production token minting • Payment Settlement: Manual Cashier entry vs. live Ghana MoMo rail

4. Recommended First Milestone & Implementation Order

Proposed Milestone: 'Verified Single-Loan State Machine Slice'
Implements the complete 1-Borrower, 1-Agency, 1-Lender workflow in Convex backend functions using integer pesewas and basis points, validated by automated tests before cloud provisioning or UI wiring.

- **Phase 1 — Schema Expansion:** Add applications, lenderProposals, and consents to frontend/convex/schema.ts.
- **Phase 2 — State Engine & Redaction:** Author lib/stateMachine.ts guards and server-side lender redaction query.
- **Phase 3 — Core Mutations:** Implement submitApplication, assignLender, submitLenderResponse, agencyApprove, and recordConsent.
- **Phase 4 — Automated Verification:** Run transition sequence and role-permission tests with npx tsc and test harness.
- **Phase 5 — Frontend Component Wiring:** Connect existing LoanApplication.tsx and SanctionViewer.tsx components.